

Geothermal gradient change during subduction recorded by ultrahigh-pressure eclogite

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The subduction of crustal rocks to, and exhumation from, depths greater than 90 km in the stability field of the mineral coesite is known as ultrahigh-pressure metamorphism, widely regarded as evidence for cold subduction. This assumption stems from peak metamorphic conditions of ultrahigh-pressure rocks, which indicate low subduction gradients of less than 330 °C GPa⁻¹, but direct evidence for such a cold prograde evolution is lacking. Here we report the discovery of a young ultrahigh-pressure eclogite in eastern Papua New Guinea. We find (sub)micrometre-sized inclusions of coesite in intermediate growth zones of garnet crystals and zircon inclusions revealing a U–Pb age of approximately 4.4 million years. Petrographic observations and thermo-barometric methods show that the prograde subduction path was initially warm with an apparent average gradient of approximately 470 °C GPa⁻¹ until a depth of about 45 km, before switching to a cold thermal regime with ultrafast subduction rates of greater than 60 mm yr⁻¹ (vertical component). This change in geothermal gradients can be explained by shear heating and/or by formation during the non-steady-state phase of a young subduction zone. We propose that high geothermal gradients at shallow levels of deep subduction systems might be common, challenging the use of peak metamorphic conditions as indicator for subduction zone thermal structures.

Subduction zones represent the connection between the exogenic and endogenic parts of Earth, enabling the effective cycling of large volumes of volatiles and thus climate stabilization over millions to billions of years¹. Most researchers agree that Earth evolved from a geodynamic regime dominated by vertical tectonics to a regime dominated by horizontal tectonics, although the timing and modes of this transition are still controversial^{2,3}. Horizontal tectonics involves subduction, and based on geothermal gradients, it is inferred that subduction evolved

from warm and shallow to cold and deep over geologic time, a trend largely interpreted as reflecting mantle cooling over Earth's history^{4,5}. Model predictions indicate that mature, modern-style subduction zones are characterized by a cold thermal structure with typical temperature–pressure (T/P) gradients <300 °C GPa⁻¹, and even the warmest subduction zones (Mexico and Cascadia) are inferred to have gradients <400 °C GPa⁻¹ (ref. 6). Thus, metamorphic rocks formed under low T/P gradients are considered indicative of modern-style subduction,

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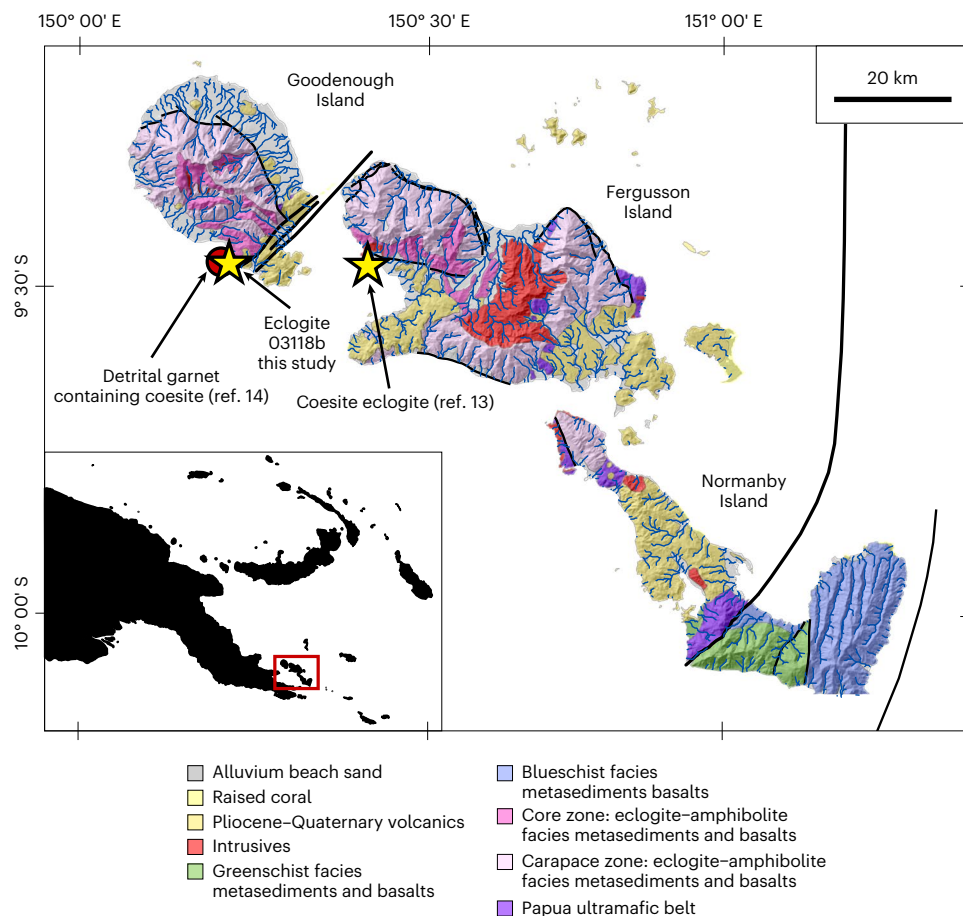


Fig. 1 | Geologic map of the D'Entrecasteaux Islands, eastern Papua New Guinea. Sample locality of the studied eclogite 03118b at Goodenough Island ($9^{\circ}28'55.84''\text{S}$, $150^{\circ}14'46.97''\text{E}$ (ref. 18)), the adjacent garnet beach placer containing garnet with coesite inclusions¹⁴ and the previously reported coesite

eclogite at Tumagabuna Island, ~400 m off the coast of Fergusson Island¹³. Black lines indicate normal and transfer faults. Inset: map of location in eastern Papua New Guinea. Figure adapted from ref. 14, National Academy of Sciences, based on the base map of ref. 11, Geological Society of America.

which includes blueschist facies and ultrahigh-pressure (UHP) rocks⁷. In blueschist facies rocks, glaucophane and/or lawsonite provide direct evidence for a cold thermal structure of the subduction zone at relatively shallow depths. In UHP rocks, however, mineralogical evidence clearly ascribed to the prograde path is generally lacking, commonly attributed to overprinting during increasing pressure and temperature. Thus, cold, prograde gradients inferred from peak pressure metamorphic conditions of UHP rocks⁸ are a necessarily simplified assumption.

Our most recent example of HP and UHP rock formation and exhumation occurs in eastern Papua New Guinea⁹, making the region ideally suited to test such assumptions. HP and UHP metamorphism resulted from the subduction of a rifted continental fragment of Australian provenance¹⁰. HP and UHP rocks are actively exhuming^{11,12}, forming topographic highs in the D'Entrecasteaux Islands (Fig. 1). Unequivocal evidence for UHP metamorphism in the form of coesite has been reported from a single eclogite at Tumagabuna Island¹³, ~400 m off the coast of Fergusson Island. Whereas further mineralogical evidence for UHP metamorphism from outcrop samples is lacking, coesite inclusions were discovered in detrital garnet from a beach placer of Goodenough Island¹⁴ (Fig. 1) using the detrital method to trace UHP metamorphism at the catchment scale^{15–17}. Besides demonstrating that UHP rocks have a larger spatial extent than previously thought, detrital garnet chemistry and mineral inclusion assemblages show that felsic and mafic rocks are involved in the UHP rock cycle¹⁴.

Here we present new data on a partially retrogressed eclogite from Goodenough Island (Fig. 1), including detailed mineral inclusion analysis in garnet, single-grain thermometry of rutile inclusions, U–Pb dating

of zircon inclusions, melt inclusion re-homogenization experiments and thermodynamic modelling. We show that the eclogite underwent UHP metamorphism under low T/P gradients in the coesite stability field, whereas the preceding garnet core growth occurred at a considerably higher T/P gradient, a condition commonly not associated with cold, modern-style subduction. We propose that the warm prograde gradient reflects shear heating and/or the early, non-steady-state phase of a young subduction zone and caution that peak metamorphic conditions alone may be insufficient to deduce the prograde evolution and geodynamic setting of UHP rock formation.

Retrogressed eclogite from Goodenough Island

Sample 03118b was collected from eastern Taleba Bay on southern Goodenough Island ($9^{\circ}28'55.84''\text{S}$, $150^{\circ}14'46.97''\text{E}$), close to the garnet beach placer studied by ref. 14 (Fig. 1). The sample is from a small lens (~90 cm × 30 cm) of mafic bulk rock composition hosted by felsic gneiss¹⁸ (Supplementary Table 1). The rock is fine grained (0.05–0.70 mm) and consists of a retrogressed eclogite mineral assemblage comprising garnet (~38 vol%) + quartz (~22 vol%) + omphacitic clinopyroxene (~16 vol%) + albitic plagioclase (~10 vol%) + tschermakitic amphibole (~10 vol%) + rutile, titanite, apatite (each ~1 vol%) + ilmenite, magnetite, biotite, epidote and sulfides (each <1 vol%). Garnet shows three distinct compositional zones with the core having the highest Ca and lowest Mg contents ($23.8 \pm 0.2\%$ grossular, $16.4 \pm 0.1\%$ pyrope), decreasing Ca and slightly increasing Mg in the mantle zone ($18.2 \pm 0.1\%$ grossular, $17.3 \pm 0.1\%$ pyrope) and increasing Ca in the rim ($20.5 \pm 0.5\%$ grossular, $17.3 \pm 0.2\%$ pyrope; Fig. 2a and Supplementary Table 1).

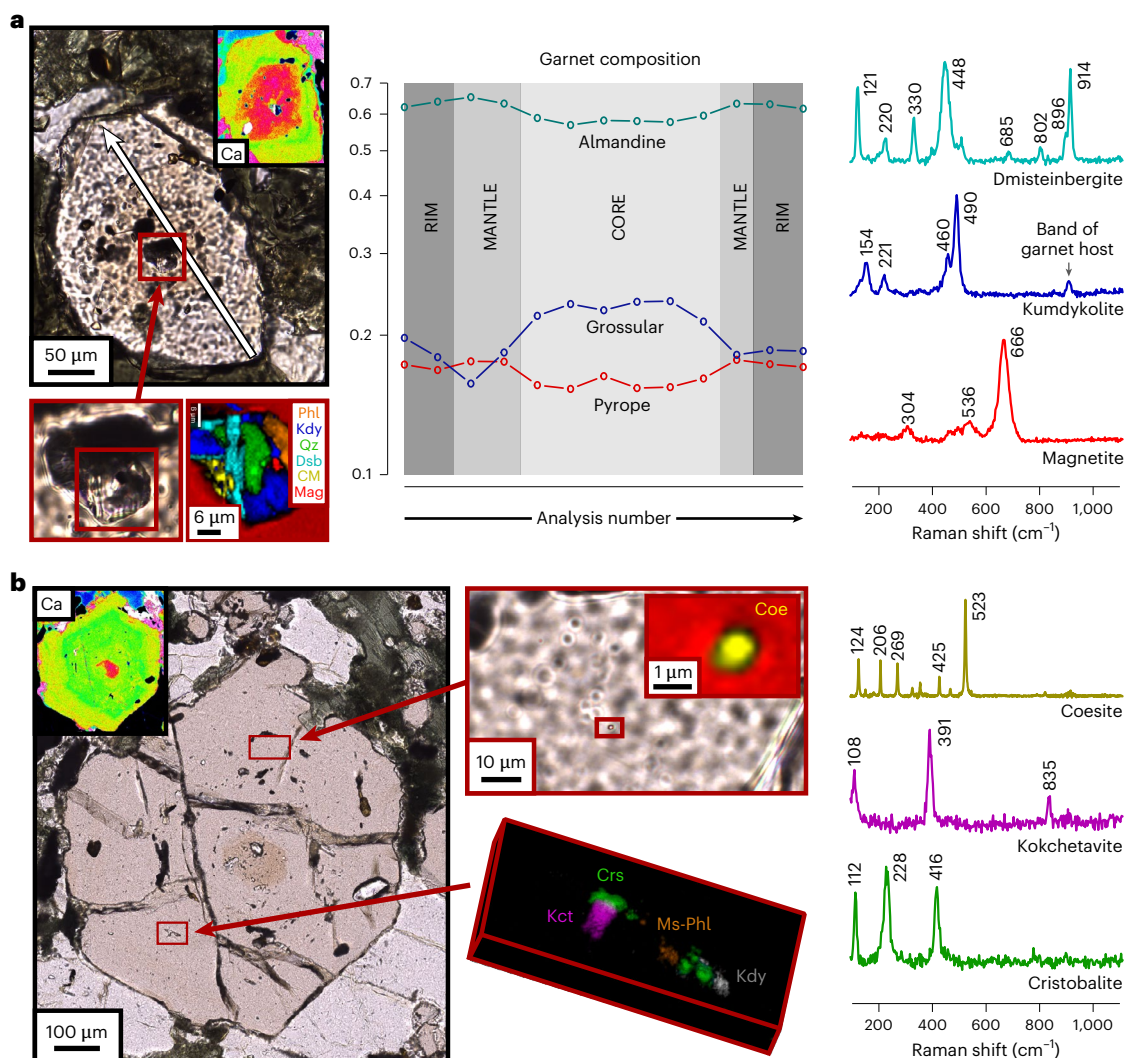


Fig. 2 | Garnet chemistry and inclusion assemblages. a, Example garnet exposing large parts of the core zone. Chemical zoning is shown by a Ca intensity map (inset) and along a profile marked by the white arrow (middle). Bottom left: detailed image shows a melt inclusion in the core zone, with mineral phases shown by a two-dimensional Raman image. Right: Raman spectra of dmisteinbergite, kumdykolite and magnetite. **b**, Example garnet exposing large parts of the mantle zone with chemical zoning shown by a Ca intensity map (inset). Middle: Raman images show a detailed two-dimensional image of an

-0.8- μm sized monomineralic coesite below the polished surface and a three-dimensional image of an irregular-shaped melt inclusion partially exposed at the polished surface. Right: Raman spectra of coesite, kokchetavite and cristobalite. See refs. 49 and 50 for the first description of kumdykolite, kokchetavite, cristobalite and dmisteinbergite in nanogranitoids. CM, carbonaceous matter; Crs, cristobalite; Dsb, dmisteinbergite; Kdy, kumdykolite; Kct, kokchetavite; Ms, muscovite; Phl, phlogopite; Qz—quartz.

Inclusions in garnet are common and most abundant in the core (Fig. 2). Garnet core, mantle and rim contain inclusions of quartz, rutile, zircon, apatite and polyphase inclusions. Inclusions of omphacite, titanite and rare ilmenite are restricted to the core, whereas inclusions of monomineralic coesite and bimineralic coesite/quartz are restricted to mantle (Fig. 2b and Extended Data Fig. 1). The larger size of the bimineralic coesite/quartz inclusions of ~5 μm compared to the monomineralic coesite inclusions of ~4–1 μm agree with the hypothesis that inclusion size is an important factor for coesite preservation¹⁹. Polyphase inclusions in core and mantle zones consist of SiO_2 polymorphs (quartz or cristobalite) + feldspar polymorphs (kumdykolite [$\text{NaAlSi}_3\text{O}_8$] and/or kokchetavite [KAlSi_3O_8] and/or dmisteinbergite [$\text{CaAl}_2\text{Si}_2\text{O}_8$]) + phyllosilicates (muscovite and/or phlogopite and/or chlorite) $\pm$ accessory magnetite and carbonaceous matter (Fig. 2). The shape of the polyphase inclusions differs between garnet core and mantle, showing isometric to negative crystal shapes in the core but highly irregular shapes in the mantle (Fig. 2a,b). Rutile inclusions in the garnet core ($n = 2$) show a

mean Zr content of $351 \pm 18 \mu\text{g g}^{-1}$, whereas rutile inclusions in the mantle ($n = 13$) show a mean content of $585 \pm 13 \mu\text{g g}^{-1}$.

Pressure–temperature conditions of garnet crystallization

Inclusions of coesite in the garnet mantle zones but exclusively quartz in the core zones provide first-order mineralogical evidence that the core formed below, and the mantle above, the quartz-to-coesite transition located at 2.7–3.0 GPa (Fig. 3). The polyphase inclusions consisting of SiO_2 polymorphs, feldspar polymorphs and phyllosilicates are compatible with so-called nanogranitoids that are melt droplets, now crystallized, trapped during garnet growth^{20–22}. Cristobalite and the feldspar polymorphs form in nanogranitoids as metastable phases due to the peculiar conditions of supersaturation and rapid crystallization experienced by a silicate melt in a small pore²³. Consequently, these polyphase inclusions indicate garnet growth in the presence of a melt, that is, at suprasolidus conditions.

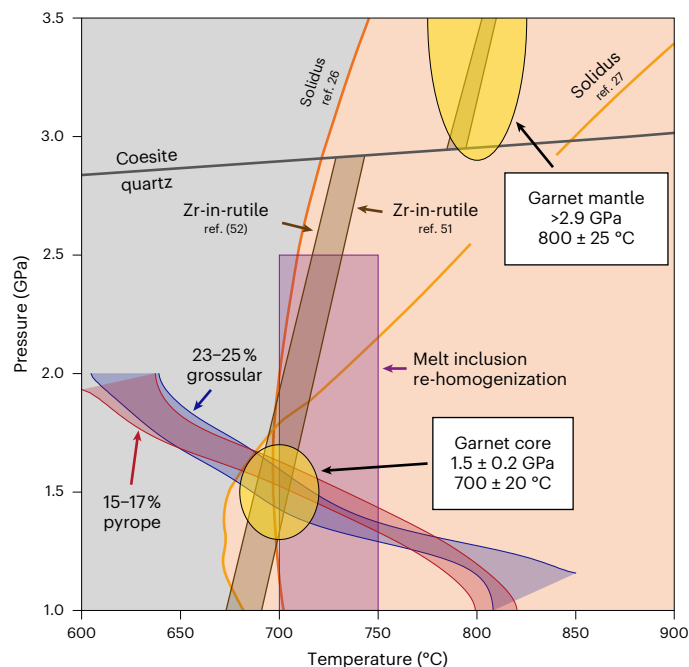


Fig. 3 | Pressure–temperature conditions of garnet core and mantle growth. Diagram shows the location of the quartz-to-coesite transition, the water-saturated solidi of basic rocks^{26,27} and grossular and pyrope isopleths for the observed garnet core composition based on thermodynamic modelling (Methods and corresponding pseudosections in Extended Data Fig. 3). Additionally shown are the results of the Zr-in-rutile thermometry for garnet core and mantle^{51,52} and the result of the re-homogenization experiment for a melt inclusion in the garnet core. Best pressure–temperature estimates for garnet core and mantle growth are indicated by yellow ellipses.

For the garnet core zone, one of these melt inclusions was successfully re-homogenized to glass at experimental conditions of 750 °C and 2.5 GPa (Extended Data Fig. 2). Previous studies^{24,25} suggest that the re-homogenization temperature of these inclusions roughly corresponds to the temperature of host garnet formation and melt entrapment, that is, the garnet core probably grew at 700–750 °C (Fig. 3, purple field), whereas the pressure represents a maximum value. The Raman spectrum of the glass shows a clear, large, asymmetric

band in the frequency region of water, suggesting the presence of an aqueous fluid during melt production (Extended Data Fig. 2). Thus, we assume a water-saturated solidus for thermodynamic modelling, met at a maximum of four mol% H₂O (Extended Data Fig. 3). Considering experimentally determined and modelling-based melting of basic rocks, temperatures >670 °C are required for partial melting^{26,27} (Fig. 3). Garnet core formation slightly above the solidus is in line with the melt re-homogenization experiment as well as Zr-in-Rutile thermometry, together indicating temperature conditions of 670–730 °C at <2.5 GPa. Compositional isopleths of the observed garnet composition cross this temperature range at ~1.5 GPa, resulting in our best estimate for garnet core formation at 1.5 ± 0.2 GPa and 700 ± 20 °C (Fig. 3). Notably, these conditions are above the thermodynamically predicted equilibrium isograd of garnet formation, which is either related to overstepping²⁸ and triggering of garnet growth by partial melting or resorption and replacement of earlier formed subsolidus garnet at suprasolidus conditions²⁹.

Coesite inclusions in the garnet mantle zone indicate pressure conditions >2.7 GPa (Fig. 2b), which is in general agreement with decreasing grossular and slightly increasing pyrope components of garnet (Extended Data Fig. 3). However, garnet compositional isopleths are insensitive to conditional changes at pressure conditions >1.8 GPa and thus do not enable conditions of garnet mantle formation to be determined. With regard to temperature, melt inclusions occurring in the garnet mantle zone show that the eclogite stayed above the solidus since garnet core formation (Fig. 2b). We note that the exact location of the solidus is uncertain at pressures >1.8 GPa (refs. 26,27) (Fig. 3). Zr-in-Rutile thermometry on rutile inclusions of the garnet mantle zone indicates temperature conditions in between the two considered solidi, resulting in our best estimate for garnet mantle formation at >2.9 GPa and 800 ± 25 °C (Fig. 3).

Following garnet mantle formation, pressure conditions decreased considering the increasing grossular and constant pyrope contents in the garnet rim (Extended Data Fig. 3). Specific conditions for garnet rim formation, however, have not been determined due to modelling limitations already observed for the mantle and also because this later stage of garnet formation is beyond the scope of this study.

U–Pb dating and composition of zircon inclusions

Whereas we did not observe zircon within the matrix of the eclogite, micrometre-sized zircon inclusions are commonly found in garnet (Extended Data Fig. 1). These were extracted by dissolving the host³⁰,

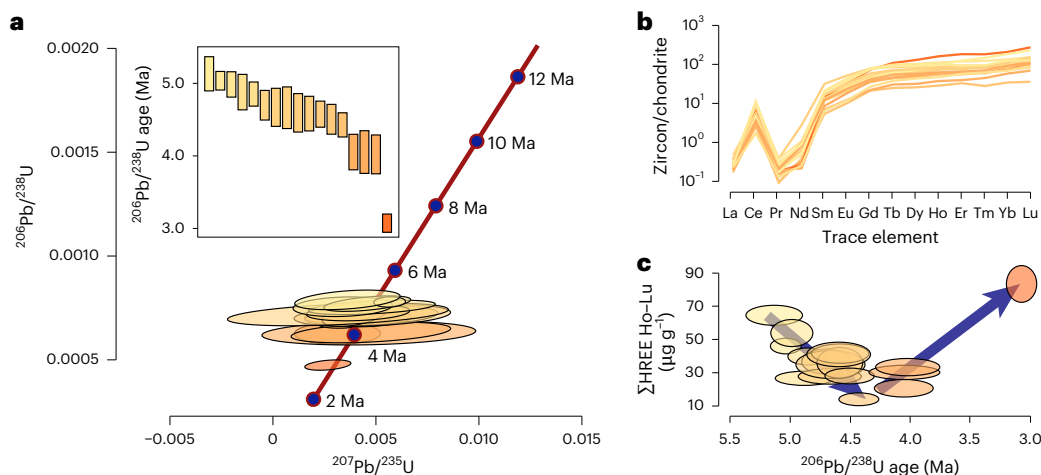


Fig. 4 | Zircon U–Pb dating and trace-element composition. **a**, Wetherill plot of filtered, ²⁰⁸Pb corrected zircon analyses ($N = 17/26$) and ordered ²⁰⁶Pb/²³⁸U ages. Inset: corresponding ²⁰⁶Pb/²³⁸U ages in chronological order. **b**, Chondrite-normalized trace-element composition of dated zircon grains. **c**, Sum of

heavy-rare-earth-element concentrations (Ho to Lu) as a function of age. All uncertainties are given as two times the standard error and ellipses, bars and lines are colour coded according to age from yellow (oldest) to orange (youngest). Supplementary Table 1 provides filtered and unfiltered data.

and 26 zircons of ~10 µm size were analysed for U–Pb isotopes and trace-element mass fractions using large geometry secondary ion mass spectroscopy (Supplementary Table 1). After excluding analyses with insufficient radiogenic Pb and those from a xenocryst, the remaining 17 dated zircon grains span an age range of 5.1 ± 0.2 million years (Ma) to 3.1 ± 0.1 Ma (Fig. 4a). Notably, these ages are older than a previously reported concordia intercept age of 2.1 ± 0.5 Ma from three zircon inclusions in garnet of the same sample¹⁸.

Trace elements of the dated zircon grains show low Th/U of <0.05, the absence of a negative Eu anomaly and a flat heavy-rare-earth-element (HREE) pattern (Fig. 4b). From 5.1 to 4.4 Ma, the sum of the concentrations of the HREEs Ho, Er, Tm, Yb and Lu decreases, followed by an increase until the youngest zircon with an age of about 3.1 Ma (Fig. 4c).

Although zircon was analysed from separates, the zircon trace-element composition can be used to link zircon dates to garnet growth zones with reasonable certainty. The absence of negative Eu anomalies in all zircon crystals shows that plagioclase was already absent during crystallization³¹ of the oldest zircon at about 5.1 Ma (Fig. 4b), which we interpret as the maximum age of garnet core growth. The flat zircon HREE patterns and the decrease in HREE until about 4.4 Ma (Fig. 4c) indicate either competition with garnet and increasing garnet content³¹ or extraction of a HREE-rich melt or a combination of both. The former is expected to be the superordinate process as garnet was present and growing during melt generation, evidenced by melt inclusions and HREE preferentially partition into garnet³². Thus, the age range of 5.1 to 4.4 Ma marks subduction from ≤ 1.5 GPa to >2.9 GPa, corresponding to vertical rates of >60 mm yr⁻¹ assuming lithostatic pressure. The age of about 4.4 Ma for the UHP stage is considerably younger than the about 7.1 Ma UHP stage³³ for the coesite eclogite at Tumaqabuna Island¹³, making it among Earth's youngest coesite eclogites. Zircon formed after about 4.4 Ma is rare and shows an increase in HREE with decreasing age (Fig. 4c), which we interpret as scarce matrix zircon crystallized during pressure reduction and garnet resorption. This potentially also explains the younger concordia intercept ages reported from the same sample¹⁸, whereby the studied zircon inclusions occur at the outer garnet margins and are connected to the matrix by fractures (compare with Fig. 6 in ref. 18). Notably, the intercept age of 2.1 ± 0.5 Ma (ref. 18) results from the assumption that the discordant analyses represent a single crystallization age, which is clearly not the case when considering the larger dataset presented here.

Geothermal gradient change during subduction

The constrained P–T conditions show that the garnet core in the UHP eclogite formed at considerably higher geothermal gradients than the garnet mantle. Whereas the garnet mantle formation conditions agree with model predictions of the thermal structure of mature, modern-style subduction zones⁶, the core formation conditions are clearly offset to higher temperatures (Fig. 5a). Notably, even within subduction zones with a young age of the downgoing slab, such as Mexico and Cascadia (~10 Ma), such models predict lower temperatures than observed for garnet core formation conditions. A potential explanation for this discrepancy is that garnet core and mantle zones developed during two different metamorphic events³⁴, but this is incompatible with the small range observed in zircon ages (5.1 to 3.1 Ma), which instead supports a single, continuous event (Fig. 4a). This also agrees with P–T paths reported from other UHP terranes that document prograde conditions^{35,36} (Fig. 5a), suggesting that UHP rocks might commonly follow high thermal gradients at shallow depth. Potential mechanisms include shear heating along the plate interface³⁷ and/or higher gradients during the early stages of subduction as a consequence of the time-dependent evolution of subduction systems³⁸, both being an inherent feature of all subduction systems.

With regard to shear heating, it has been argued that thermal models of mature subduction zones incorrectly neglect such important heat sources, based on the observation that subduction-related rocks often

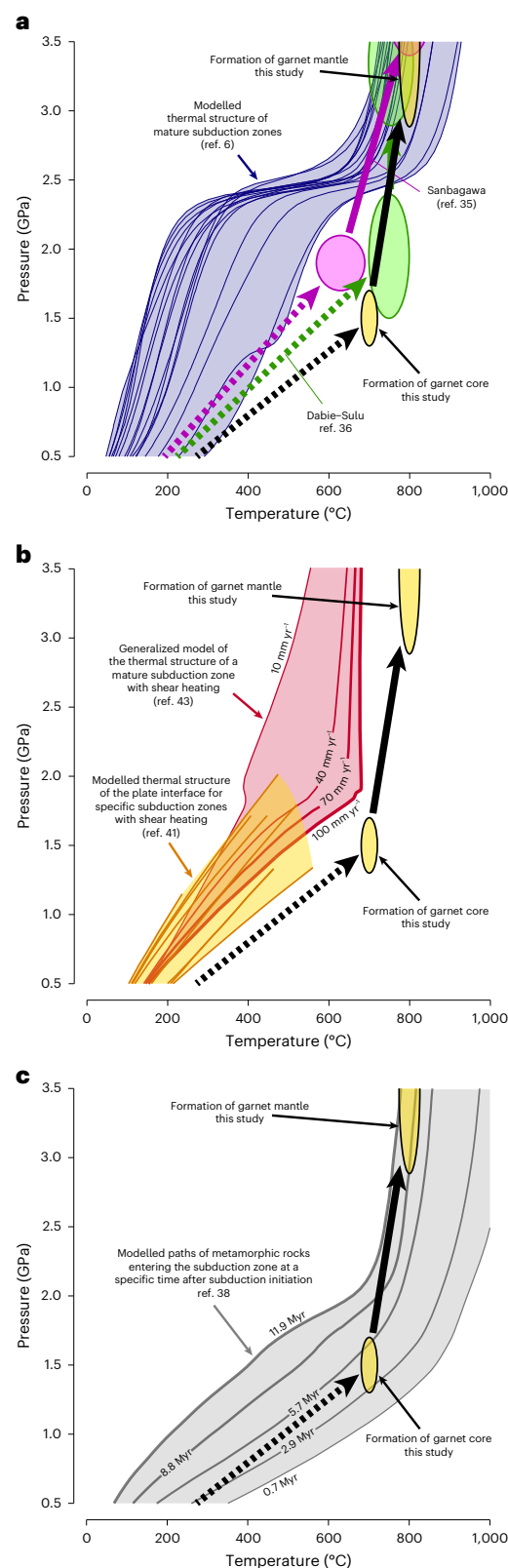


Fig. 5 | Comparing thermal subduction zone models to observed formation conditions. **a**, Thermal models of oceanic, modern, mature subduction zones⁶ (blue) compared to the data presented here and UHP rocks from Sanbagawa³⁵ and the Dabie-Sulu orogen³⁶. **b**, Thermal model of a subduction zone considering shear heating (red) with a lower plate age of 50 Ma, an effective coefficient of friction of 0.07 and for different convergence rates⁴³ and for the plate interface region of nine specific subduction zones⁴¹ (yellow). **c**, Pressure-temperature paths of rocks entering the subduction zone at a specific time after subduction initiation³⁸ (grey).

record warmer conditions than models predict^{39–41}. Considering dissipation of energy during shearing at the plate interface⁴², Fig. 5b shows the general thermal effect on the top of the subduction interface for different convergence rates⁴³ and thermal models for the plate interface region of specific subduction zones⁴¹. Compared to models without shear heating (Fig. 5a), the thermal structure more closely resembles the shape of the observed P–T path, but modelled temperatures are still lower than those observed (Fig. 5b). Nevertheless, temperatures might have been even higher if convergence rates exceeded 100 mm yr^{−1} (presently oblique convergence between the Pacific and Australian Plate occurs at a rate of ~110 mm yr^{−1} (ref. 44)), the effective coefficient of friction was unexpectedly high⁴⁵ and/or the age of the downgoing plate was young enough to become a relevant factor³⁷.

With regard to the time-dependent evolution of subduction systems, numerical models show pronounced thermal variability during the early stages of subduction before the establishment of quasi-steady-state conditions^{38,46–48}. Figure 5c shows the P–T paths of rocks entering the subduction zone at a specific time after subduction initiation, extracted from a numerical model considering time-dependence³⁸. The shape and position of the observed prograde path exhibit strong similarities to the numerically predicted paths for rocks entering the subduction zone ~5–12 Myr after subduction initiation (Fig. 5c) and the observed subduction rate of >60 mm yr^{−1} from 1.5 GPa to 2.9 GPa also agrees with model predictions for rocks entering ~8–13 Myr after initiation³⁸ (Extended Data Fig. 4). Although the observed path is slightly steeper and the modelled timescale somewhat longer, such differences are expected given the idealized, non-site-specific nature of the model.

In summary, the apparent thermal gradient recorded along the shallow prograde path (~470 °C GPa^{−1} to ~1.5 GPa) markedly exceeds that implied by the peak conditions alone (~280 °C GPa^{−1}); the rock therefore followed a warmer prograde path than peak conditions suggest. Heating due to energy dissipation during shearing and formation in a young subduction zone are both reasonable and non-mutually exclusive mechanisms to explain the high-T/P gradient observed for depths ≤45 km. While quantifying the relative contributions of the two mechanisms is beyond the scope of this study, our data clearly show that although the identification of UHP metamorphism indicates subduction to, and exhumation from, depths >90 km, it is not unequivocal evidence that subduction was exclusively cold and evolved at a thermal steady state. Consequently, considering T/P gradients of peak metamorphic conditions for reconstructing Earth's geodynamic evolution might be misleading, as some high- to intermediate-T/P rocks may have formed in subduction zones that reached cold conditions at deeper levels or at a later stage, while some low T/P rocks may have formed in subduction zones that never reached a cold, steady-state thermal structure.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41561-026-02110-1>.

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Methods

Bulk rock composition

Bulk rock composition (Supplementary Table 1) was analysed in an external facility, the Bureau Veritas Mineral Laboratories in Vancouver (Canada). The sample was analysed by inductively coupled plasma optical emission spectroscopy.

Mineral chemistry by electron microprobe analysis

Electron microprobe analysis was performed to determine mineral chemistry (Supplementary Table 1). Major oxides in garnet, pyroxene, amphibole and plagioclase were analysed at the Institute of Geosciences, University of Potsdam (Germany) using a JEOL JXA-8200 microprobe. Measurements were performed with an accelerating voltage of 15 kV, a beam current of 15 nA and a probe diameter of 2 μm . Counting times were 20 s on peak and 10 s on background, except for K, Na and Cl with 10-s counting time on peak and 5 s on background. The Zr content in rutile inclusions in garnet was analysed at the Goettingen laboratory for correlative Light and Electron Microscopy (GoeLEM) using a JEOL JXA iHP200F electron microprobe. Measurements were performed using a focused beam at the inclusion centre with an accelerating voltage of 20 kV and a beam current of 120 nA. Counting times for Zr were 200 s on peak and 100 s on background. All uncertainties are reported as standard error of the mean.

Inclusion assemblage investigation by Raman spectroscopy

Raman spectroscopy was performed at the Geosciences Center Göttingen (Germany). The experimental set-up includes a WITec alpha 300 R fibre-coupled ultrahigh throughput Raman spectrometer, a 532-nm laser, an automatically adjusted laser power of 30 mW, a 100 $\times$ long working distance objective with a numerical aperture of 0.75 and an electron multiplying charge-coupled device. Two-dimensional images of coesite inclusions were acquired using a 1,200 lines $\times$ mm⁻¹ grating, a step size of 0.20 $\mu\text{m} \times$ 0.20 μm (Fig. 2b) and 0.25 $\mu\text{m} \times$ 0.25 μm (Extended Data Fig. 1) and an acquisition time of 1.0 s per spectrum. The two-dimensional melt inclusion image was acquired using a 300 lines $\times$ mm⁻¹ grating, a step size of 0.50 $\mu\text{m} \times$ 0.50 μm and an acquisition time of 0.5 s (Fig. 2a). The three-dimensional melt inclusion image was acquired using a 300 lines $\times$ mm⁻¹ grating, a step size of 0.25 $\mu\text{m} \times$ 0.25 $\mu\text{m} \times$ 0.50 μm and an acquisition time of 0.1 s (Fig. 2b).

Additionally, a pre-investigation of inclusion assemblages and the analysis of the glass in the re-homogenized melt inclusion (Extended Data Fig. 2) were performed at the Institute of Geosciences, University of Potsdam (Germany) using a LabRAM HR 800 (HORIBA Jobin Yvon) Raman spectrometer, a 532 nm laser and a 300 lines $\times$ mm⁻¹ grating. Acquisition time was adjusted between 10–30 s (two accumulations) to result in a good signal noise ratio.

Melt inclusion re-homogenization experiments

To verify that the polyphase inclusions were originally droplets of melt, multiple garnet chips containing crack-free, unexposed nanogranitoids were mechanically separated from at least 30 garnet crystals extracted from two different double-polished thick sections. These chips were used as starting material for our experiments using a piston cylinder apparatus (type: Boyle and England) available at GeoForschungsZentrum (GFZ) Potsdam, following the approach described in ref. 21: ‘nanogranitoid-bearing chips of garnet were loaded in platinum capsules (4 mm long and with a diameter of 3 mm) together with quartz powder to separate the garnet chips from each other and from the capsule walls. The capsules were cold sealed and embedded in crushable alumina (Al₂O₃) and then in a talc–pyrex–graphite assemblage. The experiments involved two steps: first compression and then heating under dry conditions for 24 h. Pressure calibration relied on the quartz–coesite transition with an accuracy of ± 0.05 GPa. Temperature was controlled with a K-type thermocouple (Ni = CrNi), providing an accuracy of ± 10 °C. Each experiment was then quenched, the capsule

mounted in epoxy and polished to expose the re-homogenized inclusions within the garnet chips’.

The experimental P–T conditions were chosen based on independent constraints from previous studies. The aim was to obtain a complete homogenization to glass with no host–inclusion interaction and formation of new phases or shrinkage bubbles within the inclusions (a detailed discussion of this procedure is available in refs. 24 and 53). Pressure was set to 2.5 GPa, whereas experiments were run at three different temperatures, 750 °C, 800 °C, 900 °C. The investigation of the experimental products revealed that the experiments at 750 °C were the only successful ones in re-homogenizing the inclusions without producing melt–garnet interaction (Extended Data Fig. 2), interpreted in previous studies as indicative of overheating, that is, beyond the likely temperature of entrapment of the melt in garnet, during the experimental run.

Thermodynamic modelling

The equilibrium phase diagram (Extended Data Fig. 3) was calculated using Perple_X 7.1.12⁵⁴ (2025 update) and an updated version of the internally consistent dataset⁵⁵ (ds63, 2017 update). The activity composition (a–x) models are from ref. 56 for garnet, biotite, chlorite, mica and ilmenite; ref. 57 for ternary feldspar; ref. 55 for epidote; ref. 58 for omphacitic pyroxene and ref. 27 for clin amphibole and tonalitic melt. Quartz, coesite, rutile and aqueous fluid (H₂O) are assumed as pure phases. Calculations were performed by transforming the analysed bulk rock composition (Supplementary Table 1) into the simplified model system Na₂O–CaO–K₂O–FeO–MgO–Al₂O₃–SiO₂–H₂O–TiO₂–O₂ (NCKFMASHTO). The presence of accessory hematite, magnetite and sulfides indicates slightly oxidative conditions only and thus, the amount of O was set to convert 10% of total iron to Fe₂O₃. On the basis of the large amounts of water observed in the re-homogenized melt inclusions (Extended Data Fig. 2), the amount of H₂O was set to saturate the solidus between 1.0 and 3.5 GPa, which is reached at a maximum of four mol%.

Zircon U–Pb dating and trace-element composition

As zircon was not observed in the matrix of sample 03118b but as small inclusions in garnet, zircon was separated using a slightly modified protocol of the bulk-annealing- and dissolution-based concentration method³⁰. The sample was crushed to <200 μm using a jaw crusher and heated to 900 °C for 60 h before dissolution. Dissolution was performed in four steps, each in a Teflon vessel on a heating plate at -120 °C, using a sample-to-acid ratio of 1:10, followed by four times of decanting and rinsing with MilliQ water. In the first step, aqua regia (37% HCl and 65% HNO₃ at a ratio of 3:1) was given to the sample for 24 h, followed by hydrofluoric acid (48%) for 24 h, followed by aqua regia for 6 h and finally hydrochloric acid (37%) and boric acid (6%) at a ratio of 1:1 for 4 h. This resulted in a concentrate strongly dominated by rutile with minor proportions of small zircon (mostly ≤ 10 μm) and accessory aluminosilicates. The largest zircon grains with a size of -10 μm were handpicked under the stereo microscope and pressed into high-purity aluminium (5 N) at 3,000 N for one minute. After cleaning with Decon solution and rinsing with deionized water and isopropanol, the mount was coated with 80-nm gold.

Zircon U–Pb isotopes and trace-element mass fractions were analysed using a CAMECA IMS 1300HR3 large geometry secondary ion mass spectrometer at the John de Laeter Centre, Curtin University, following instrumental conditions described in ref. 59. For U–Pb isotopes, a 12 nA ¹⁶O⁻ primary beam in Gaussian mode was focused to an -10- μm spot. Temora-2⁶⁰ was used as primary reference material and 91500⁶¹ as secondary reference material. The secondary reference material age of 1,065 Ma was well reproduced with $1,063 \pm 9$ Ma ($n = 5$, two times the standard error). For trace elements, a 10 nA ¹⁶O⁻ primary beam in Gaussian mode was focused to an -10- μm spot. Standard Reference Material 610 of the National Institute of Standards and

Technology (NIST SRM 610) was used as primary reference material with the preferred values from ref. 62, and 91500^{63,64} was analysed in the same run as a secondary reference material. Repeated analyses yielded close agreement between measured (with two times the standard error for $n = 5$) and reference values (in parentheses) for key trace-element ratios: $\text{Eu}/\text{Eu}^* = 0.70 \pm 0.10$ (0.72), $\text{Yb}/\text{Gd} = 58.4 \pm 3.8$ (55.3) and $\text{Th}/\text{U} = 0.31 \pm 0.02$ (0.34).

From the total of 26 analysed grains, eight analyses were excluded due to radiogenic ^{206}Pb proportions $<90\%$. In addition, one (probably xenocrystic) grain giving an age of 362 ± 6 Ma was excluded. The remaining 17 grains show low Th concentrations with $\text{Th}/\text{U} < 0.05$. Thus, a ^{208}Pb correction was applied using the modern Pb composition⁶⁵. Data are provided in Supplementary Table 1. All uncertainties of individual analyses are reported as two times the standard error.

Data availability

The data for bulk rock composition (Inductively Coupled Plasma Optical Emission Spectroscopy), mineral chemistry (Electron Probe Micro Analysis) and zircon U–Pb isotopes and trace-element composition (Secondary Ion Mass Spectrometry) are provided as Supplementary Table 1 and are available via Figshare at <https://doi.org/10.6084/m9.figshare.33306585> (ref. 66).

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Author contributions

J.S., S.F. and S.L.B. designed the research. S.L.B. sampled the investigated rock. J.S. and K.G. performed electron microprobe analysis. J.S., S.F. and K.G. performed Raman spectroscopy analysis. S.F. selected material for melt inclusion re-homogenization experiments, and B.W. performed the experiments. D.S. performed thermodynamic modelling. J.S. extracted zircon inclusions, and A.K.S. performed ion microprobe analysis. A.F.H. contributed model data on the temporal evolution of the subduction zone thermal structure. J.S., S.F., K.G., A.K.S., B.W. and A.F.H. analysed and evaluated the data. J.S., S.F., D.S. and S.L.B. visualized the data. J.S. wrote the initial paper, and all authors reviewed the paper.

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Competing interests

The authors declare no competing interests.

Additional information

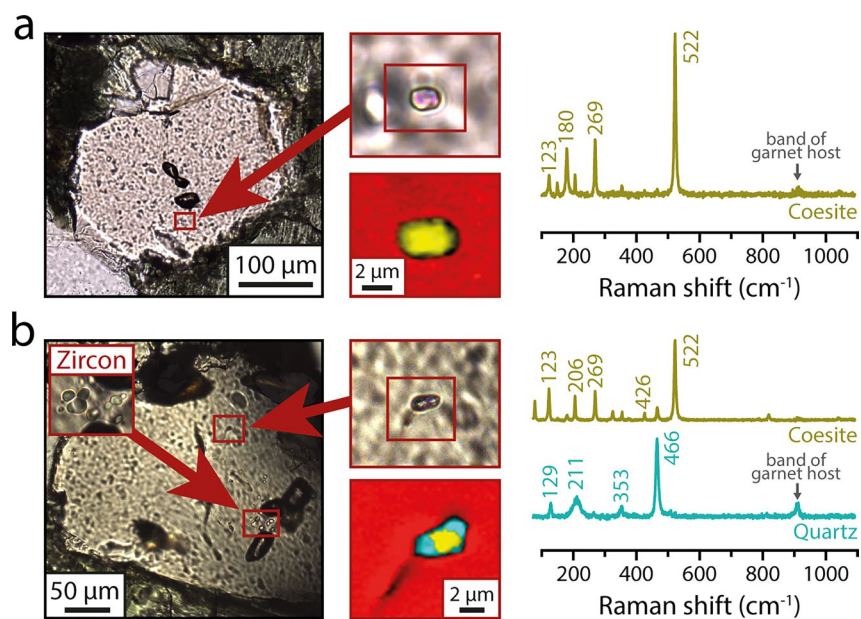
Extended data is available for this paper at <https://doi.org/10.1038/s41561-026-02110-1>.

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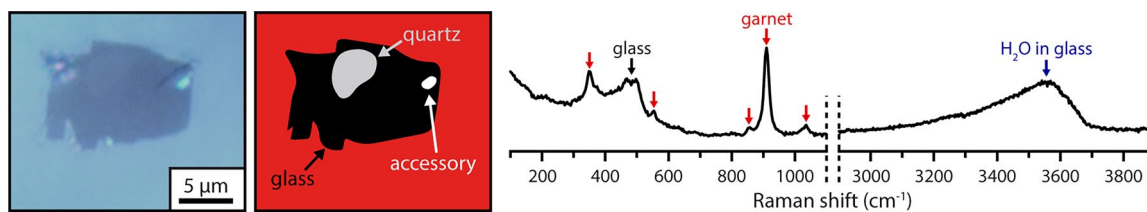
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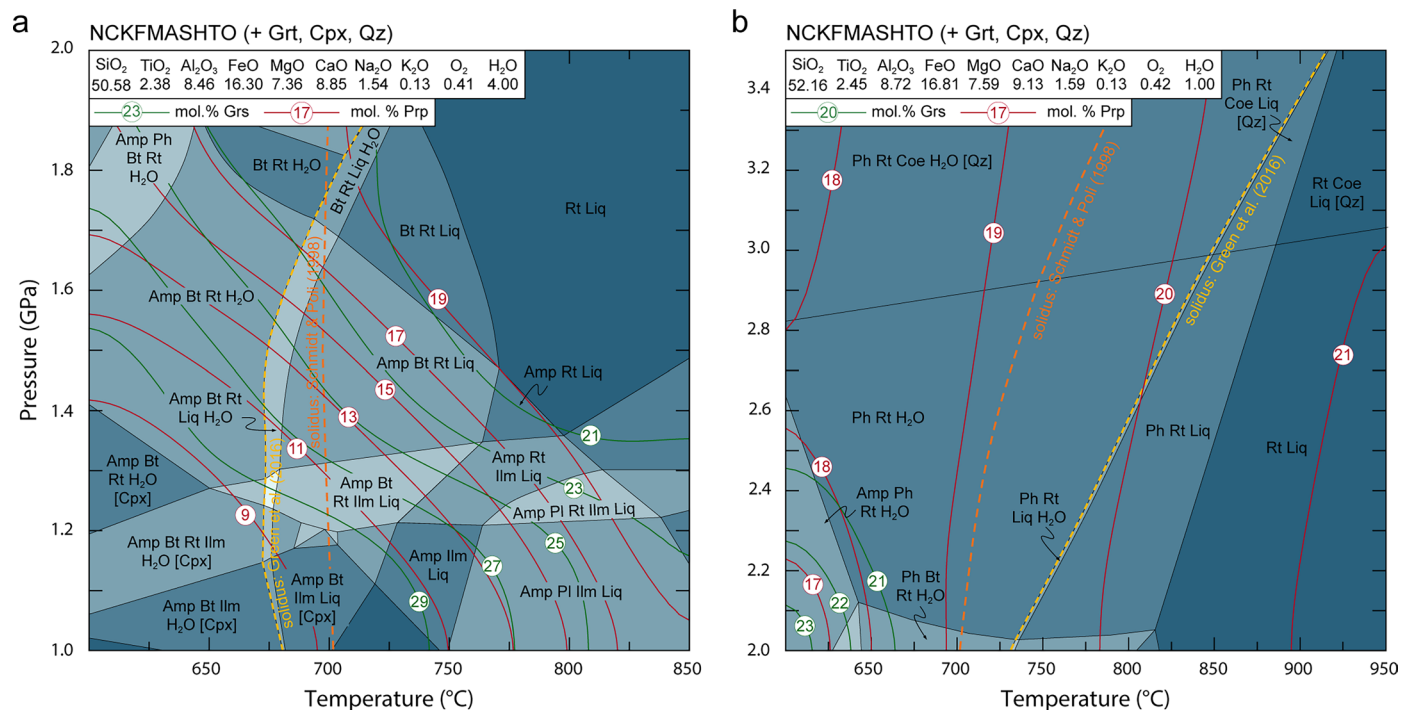
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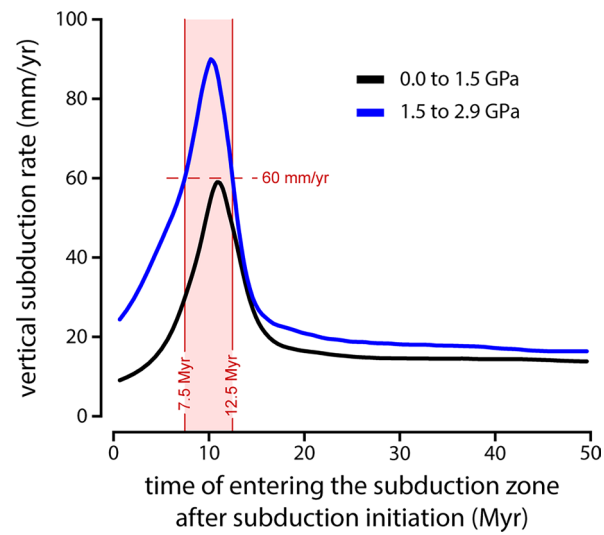
Extended Data Fig. 1 | Further examples of coesite inclusions in garnet mantle zones. a, monomineralic coesite inclusion. **b,** bimineralic coesite/quartz inclusion showing fracturing of the garnet host. Inset shows zircon inclusions.



Extended Data Fig. 2 | Melt inclusion in garnet core zone re-homogenized to glass. Shown are a reflected light image, a schematic illustration of the glass together with accidentally trapped quartz and an accessory phase, and the Raman spectrum.



Extended Data Fig. 3 | Equilibrium phase diagrams of sample 03118b. a, phase diagram for garnet core formation with 4 mol% H₂O to saturate the solidus. **b**, phase diagram for garnet mantle formation with 1 mol% H₂O to saturate the solidus.



Extended Data Fig. 4 | Vertical subduction rates according to the time-dependent evolution model after ref. 38. Modelled subduction rates from 1.5 GPa to 3.0 GPa are > 60 mm/yr in the vertical components for rocks entering the subduction zone between 7.5 Myr and 12.5 Myr after subduction initiation.